

Privacy Concerns regarding Google Privacy and Intellectual Property Rights

GÁBOR VITRAI - MAT. 230360, University of Trento, Italy

This document was created for the Privacy and Intellectual Property Rights subject as a final - end of semester - assignment. The paper aims to write about Privacy Concerns regarding Google. The paper tries to collect as many stories as possible about concerns, lawsuits, cases from both when governments/individuals were right, and both when Google won.

Additional Key Words and Phrases: Privacy Concern, Google, Data Protection, Regulations

ACM Reference Format:

Gábor Vitrai - MAT. 230360. 2022. Privacy Concerns regarding Google Privacy and Intellectual Property Rights . 1, 1 (February 2022), 7 pages. <https://doi.org/10.1145/nnnnnnn.nnnnnnn>

1 CONCERNS WITH GOOGLE

In this section, I aim to present how Google collects data, what they use it for and what methods are they using to collect data. In this section I will write about different topics which are related to concerns, however I will also write about how google aims to use these services / tools to enhance user experience.

1.1 The initial problem

Upon creating a Google account, we get access to countless services. Google Maps, Calendar, Drive, News, Chrome, Google Photos, etc... Each of them brings their own questions to the topic. Will the Google maps know my location? Is it going to be saved? Is the company going to use my daily routine for recommendations? Are they using my data in the cloud? Lets see some real life example.

Author's address: Gábor Vitrai - MAT. 230360, gabor.vitrai@studenti.unitn.it, University of Trento, Trento, Trentino, Italy, 38123.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2022 Association for Computing Machinery.

XXXX-XXXX/2022/2-ART \$15.00

<https://doi.org/10.1145/nnnnnnn.nnnnnnn>

1.2 Data Leaks

Every tech giant has its own fight with hackers, software errors or human negligence. Companies storing large amount of data, can easily become target of attacks, thus having a lot of employees also increases the chance of human error. For example, In 2009, Google reported a bug in Google Docs (word like online Office service) had allowed unintended access to some private documents. [1]. While, only 0.05% of the stored documents were stored, in many cases the leaked data is valuable.

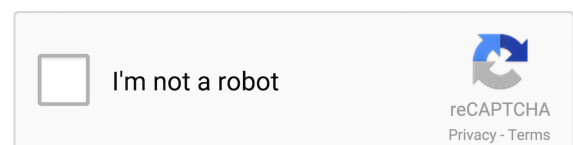
1.3 Tracking

Tracking can be divided into two main groups. Location tracking and internet tracking. Location Tracking. Timeline, why is it good: summaries, recommendations form history upon searching for a place.

Following the route of a user on the internet is not the most ethical thing to do, however, it is crucial for Google. This kind of data is invaluable for Google itself to increase the efficiency of their own marketing/advertising abilities. Because of the usefulness of this data, Google is constantly suspected of collecting information about Internet users through various tools, provided by Google to developers. These tools for example are Google Analytics itself, Google Play Services or reCAPTCHA (Which is a tool used to prove that the user is not a robot).

The important point is that, for example in the case of reCAPTCHA we actually don't know how can it decide if the user is a robot or not. We do not know, - it is not public - how the tool works, what factors does it decide on. "Google is analysing the behaviour before, during and after clicking the checkbox to determine whether a user is human. This analysis **might** include everything from your browsing history, to the way the user organically move the mouse on the page. [2]

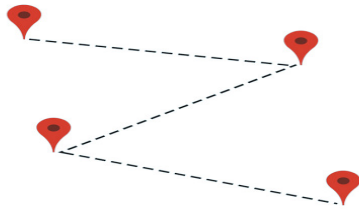
Fig. 1. Google's reCAPTCHA



"ReCAPTCHA uses the google.com domain instead of one specific to reCAPTCHA. This allows Google to receive any cookies that they have already set for the user, effectively bypassing restrictions on setting third party cookies and allowing traffic correlation with all of Google's other services, which most users use. ReCAPTCHA collects enough information that it could reliably de-anonymize many users that simply wish to prove that they are not a robot" [3]

1.3.1 Appealing Users: Following the continuous lawsuits over important tracking and unknown data retention periods, Google tried to appeal to the increasing number of privacy conscious people. At Google IO 2019 [4], they announced plans to limit the data retention period **for some of its services**. Users could choose between 3 to 18 months. Now, in 2022, this feature is implemented and available to users. This setting regarding how long Google can "store" these information now include google maps and other highly used applications, which are collecting data of users. However, still, this data retention limit is disabled by default. [3]

Fig. 2. Tracking of User's internet history



1.4 Cookies

It is not a new information that Google uses cookies just like many other online providers. Many years ago cookies by Google also had privacy concerns. The cookie used to Access the Google accounts, originally didn't expire for 32 years. That was obviously a huge problem, as these can be used to track user history, even on unrelated websites. In 2007 Google reduced this length to 2 years, and in 2011, Google introduced a solution which anonymizes the IP address it uses after 9 months. As of 2016, Google's privacy policy does not promise anything about when the records about the user's searching history is deleted from its databases. [5]

In the case of Google Buzz (google's attempt for a Facebook like service) Gmail users had their contact lists immediately made public unless they manually opted out. Google shares this information with law enforcement and other government agencies upon receiving a request. The majority of these requests do not involve review or approval by any court or judge. [6]

1.5 Gmail

Whatever the topic is, google states that it's services only use the someone's data to personalize services and advertisements. The situation is the same in the case of the Gmail

service. Google claims that mail sent to or from Gmail is never read or accessed by anyone other than the holder of the account, and content read by computers is only used to improve the advertisements and the spam blocking service. The company had multiple lawsuits and open letters attacking it's privacy policies and the way how they handle a user's "property". In 2004 civil liberty organizations wrote a letter in this topic, while in 2013, Microsoft did a campaign to attach Google for scanning the content of email messages. Microsoft stated that their mailing service does not scan the content of digital letters. [7]

Other concerns include the unlimited period for data storage policies allow, and the potential for **unintended** secondary uses of the information Gmail stores.

1.6 CIA & NSA

The relationship between google or one of the Security offices was always unclear. For example in 2010, Google was reported to be working on an agreement with the NSA (National Security Agency) to investigate some cyber attacks, and while Google didn't give access to to user data (emails, communications) concerns were made due to the agreement. [10]

Fig. 3. NSA Logo



An other case was in 2013 when The Guardian included Google on a list of companies which were cooperating with one of the NSA's surveillance program based on leaked NSA documents. Following the leak, governments acknowledged the existence of the program and according to the leaked documents, the NSA has direct access to servers of those companies. As always, Google has denied the existence of any backdoor for the governments. [11]

1.7 Google Chrome

The next in our list is again data collection by Google using its own web browser, Google Chrome. In 2008 a video was uploaded to the internet showing that Google Chrome records what a user types into the URL bar. This information was sent to the Google servers to increase the numbers of search suggestions. Yet again, google stated that they do not use this information for anything else, but this event opened basis for many arguments regarding the potential privacy implications of this feature. [12]

1.8 Incognito Mode

To continue with the topic of the browser, a very common doubt raising concern of everyday users is the actual functionality of the incognito browsing mode in Google Chrome. This feature should prevent the browser (and all the Google services) from permanently store any browsing or download history, including cookies. However, websites visited, can still track and store information about the visits. In practice, any of the searches executed while being signed into a Google account will be stored as part of the account's web history. (As I wrote about this earlier in the cookies subsection, the duration of storing this information now can be limited). [13]

1.9 Google Street View

An other well known problem regarding privacy is Google's online map service, the Street View. This service has been accused of taking pictures of people's private homes (area like gardens), while an other accuse is the fact, that not all of the important details are blurred on the photos, (like license plates) as this is very challenging and computationally demanding. The 3rd concern right always is that people on the street get photographed when they do not know they are being photographed. The question arises: Who allowed google to take photos of everyone without the 100% testimony that everything what is said to be sensitive, will be blurred? [14] Other cases regarding this service of Google, will be shown later in the next section.

Fig. 4. Example of Agreement Solutions



1.10 YouTube

In 2008, Viacom (It was an American multinational mass media conglomerate with interests primarily in film and television) compromised to protect YouTube users' personal data in their one billion copyright lawsuit. Google agreed, they will anonymize user information and internet protocol addresses from its subsidiary before handing the data over to the tech giant. YouTube was forced to pay \$170 million and **implement new privacy systems**. This time google made the mandatory step.

1.11 Google's case with Wi-Fi

For 4 years, Google Street View camera cars collected about 0,6 Terabyte of data from users of not secured public and private Wi-Fi networks. No disclosures whatsoever was given to those who were "harmed".

Google apologized and said that they were "aware that they totally failed" in terms of privacy protection and that they were not aware of the problem until an inquiry from German regulators was received [24], and that none of the private data was used in Google's search engine or other services.

The Street View data collection prompted several lawsuits in the United State. I will write about in more detail later in the paper.

Currently Google no longer collects Wi-Fi data via Street View, instead using an Android device's Wi-Fi positioning system. On the other had they recommended to create a unified solution for opting-out from Wi-Fi-based positioning systems.

1.12 Do not Track

In 2011, Google was criticized for not signing onto the Do Not Track feature for Chrome that was being incorporated in most other modern web browsers. [22] It used to be an official HTTP header field, designed to allow internet users to opt-out of tracking by websites. Mozilla developer Asa Dotzler noted: "It seems pretty obvious to me that the Chrome team is bowing to pressure from Google's advertising business and that's a real shame. I had **hoped** they'd demonstrate a **bitmore independence than that**." [23]

"At the time of the criticisms, Google argued that the technology is useless, as advertisers are not required to obey the user's tracking preferences. As an alternative, Google offered an other extension called "Keep My Opt-Outs" that prevents advertising companies from **installing** cookies on the computer." [25]

Yet again, this gave doubt for users and companies if google really stops giving recommended, personalized advertisements, and stops tracking of user's history, then why do they provide their own extension instead of going with the incorporated solution.

Fig. 5. HTTP Header Marker



2 PRIVACY AND DATA PROTECTION CASES AND ISSUES BY COUNTRY

2.1 EU - Which laws and regulations are involved?

Fig. 6. European Union



2.1.1 Data Protection Directive. The Data Protection Directive, officially Directive 95/46/EC, is a European Union directive which regulates the processing of personal data within the European Union and the free movement of such data. The Data Protection Directive is an important component of **EU privacy and human rights law**.

The principles set out in the Data Protection Directive are aimed at the **protection of fundamental rights and freedom in the processing of personal data**. The General Data Protection Regulation, adopted in April 2016, superseded the Data Protection Directive and became enforceable on 25 May 2018. [15]

2.1.2 When Google listened to the EU. In 2007 the party who worked on the Data Protection Directive, wrote to Google asking the them to justify their policy of keeping information on users. They wanted to maximize the interval for 2 years. The letter doubted if the company has really fulfilled all the requirements on the EU laws regarding data protection. In the same year Google agreed that their privacy policy was not perfect and they are working on the issue.

2.1.3 Unfinished affairs of the EU and Google. In 2012, when the company merged its policies together into a big one, the same EU group assessed that it failed to comply with the EU legal framework. As a result, several countries started to investigate for possible breaches of their privacy.

2.1.4 Google Spain v AEPD. is a decision by the Court of Justice of the European Union. It held that an Internet search engine operator is **responsible for the processing** that it carries out **of personal information** which appears on web pages published by third parties. This was a case partly at Spain's national court which required Google to comply with the EU privacy laws.

2.2 France

8 years ago, a French authority, sanctioned Google, requiring it to pay their highest fee and to display on its search engine website a banner referring to the decision. Google actually complied. [17]

2.3 Germany

Here is a case where google actually stopped one of it's service despite him being legal. In 2010, vandals in Germany targeted houses that had opted out of Google's Street View. In 2011, Google decided not to expand its Street View program in Germany, but what has already been photographed—around 20 cities will remain available. This decision came despite an earlier Berlin State Supreme Court ruling that **Google's Street View program was legal**.

2.3.1 Google Being late. In 2010, Google was not able to finish within a deadline made by a German data protection supervisor to hand over data **illegally collected** from unsecured home wireless networks. (As mentioned in an anowe section). Google apologized and a month later the data was turned over to German, French, and Spanish authorities. [16]

2.3.2 Open Internet Project. Afterwards, a number of French and German companies came together to form a group called the seeking a ban of Google's manipulative favoring of its own services and content over those of others. [17]

Fig. 7. Open Internet Project



2.4 Italy

Also in 2010, there was a complaint brought by an Italian group for people with Down's Syndrome. Three Google executives were handed 6 month suspended sentences for breach of the Italian Personal Data Protection Code due to a video uploaded to the service. The video showed a disabled boy being bullied by classmates. Two years later the case was eventually dismissed.

2.5 UK

Since 2015, the Court of Appeal ruled that British people have the right to sue google in the UK for misuse of private information.[18]

2.6 Czech Republic

In 2010 The Czech government and Google, could not agree after 5 months of negotiations. The point of the Czech Republic was that Google Street View "goes beyond the extent of the ordinary sight from a street", and claimed that it "disproportionately invaded citizens' privacy".

This Lasted till 2012, when the two sides managed to agree to a number of limitations, similar to concessions Google has made in other countries. This also shows, that Google and its privacy department really works to make it's services available to everyone, independently of countries.

2.7 United States

Fig. 8. United States of America



2.7.1 Minor attempt: In 2008, a Pittsburgh couple, sued Google for "invasion of privacy". They stated that Street View made a photo of their house which is now can be found on the internet, and as a consequence it reduced the value of their building. They lost their case in a Pennsylvania court. The family was paid **one** dollar by Google for the incident.

2.7.2 Apple's Privacy Setting. In 2012-13, Google reached two settlements over tracking people online, without their knowledge after bypassing privacy settings in Apple's Safari browser. In total, Google was fined 39,5 million US dollars. Google denied doing anything wrong, but stated that they discontinued circumventing the settings early in 2012, after the case was publicly reported, and stopped tracking Safari users and showing them **personalized ads**.

2.7.3 A Case regarding young age. In 2019, Google was fined \$170 million by the Federal Trade Commission in New York for violation of COPPA regulations on its YouTube platform. (Children's Online Privacy Protection Act) [19] YouTube was required to add a feature to allow video up-loaders to flag videos made for users under 13 years old. Adverts were prohibited in videos flagged as such.

3 LEGAL QUESTIONS

3.1 How is Google (supposedly) able to overcome legal limits?

As a general look of what we see, Google doesn't go around laws, they do not cheat with laws, however they have a huge amount of diverse services, and here and there they do exceed the legally allowed limits. The company's situation is not easy as the regulations can be different in every country. For this, google has teams to work out governmental agreements with the smallest trade offs possible.

To sum up, google really tries it's best to find a workaround for legal issues, however new services and features (see the example of Google Street View) sometimes hit legal barriers, where they have to give up, either for governmental pressure, or for the will and comfort of citizens.

3.2 Is google really limited?

The Example of Double Click ads. Quite recently, 6 years ago, Google quietly dropped its ban on personally-identifiable info

Table 1. Data Requests by Country

Country	Data Requests	Country	Removal Request
1. Brazil	3663	1. Brazil	291
2. US	3580	2. Germany	188
3. UK	1166	3. India	142
4. India	1061	4. US	123

in its DoubleClick ad service. DoubleClick was an advertising company that developed and provided Internet ad services since 1995, until Google bought it in 2008. [20] Google's privacy policy was changed to state it "may" combine web-browsing records obtained through DoubleClick. New users were automatically opted in, existing users were asked if they wanted to opt in. Google contacted ProPublica, an company with Investigative Journalism to correct the fact that it didn't "currently" use Gmail keywords to target web ads. [21]

3.3 How do countries face these issues?

3.3.1 Governmental Requests: Google has been criticized for disclosing all sorts of information from governments and countries. In 2010, Google released details about how often countries around the world ask it to hand over user data or to censor information. Google declared that the majority of these requests are valid, reasonable and the information needed is for legitimate criminal investigations or for the removal of child pornography. [8]

Between July and December 2009 the data requests were the following:

3.3.2 China: Google stopped offering search services in China a month before this data was released, said it could not release information on requests from the Chinese government. [8]

3.3.3 US: In 2019, quite recently, the U.S Supreme Court risked an 8.5 million settlement which Google formed to fix a lawsuit with the claims of invading their privacy. [9]

Fig. 9. US Government



4 CONCLUSION

These articles are quire old and the number of problems regarding Google are getting lower by every year. This shows

that Google really tries to reduce concerns. Google I/O events and announcements support this as well. Google provides a very wide range of services, and says to use all the collected data to enhance user experience and to improve the services. Can we really notice the improvements? Yes! An everyday user can really experience how the search engine is getting better, how "Street View" service helps getting along in our every day life. This year, text completion was also introduced upon writing Gmail letters, and many more features are constantly being added into the variety of services.

Fig. 10. Google Services



5 APPENDIX: COOPERATION OF GOOGLE & APPLE

In 2020, a month after the outbreak of COVID-19, Google and Apple announced a partnership to take research into undertaking Bluetooth-powered Covid-19 contact tracing in a privacy preserving manner. This solution would contain no central database, and it would be available as a tool inside the OS of each operating system provider's phone. (Android and IOS) Beforehand, many startups and multinational companies (including governmental orders as well) tried to create applications for this purpose. However all the applications / solutions were doomed to fail. This is due to the power intensive need of the Bluetooth technology itself, thus these solutions were crippled by the same baked-in Bluetooth restrictions that stop normal apps to secretly track the user.

The contact-tracing toolkit has been both praised and supported. Its “decentralised” (blockchain) approach, with no sensitive central database of who-saw-who, has been supported by hundreds of privacy, security and human rights scholars. Through a software update, Apple and Google loosened privacy restrictions enough to allow public health authorities to run decentralised contact-tracing apps, but did not engineer new functionality to let apps send the unique Bluetooth identities of phones they encountered to a central server. Data had to remain secretly on phones, which is fine. Tensions grew bigger as it became 100% clear that the firms did not intend to achieve a further global change to their operating systems to specifically accommodate these countries. As a result the story of the centralised app died. NHSX (The United Kingdom’s Government unit with responsibility for setting national policy and developing best practice for National Health Service (NHS) technology) piloted an app relying on fragile workarounds to avoid the privacy restrictions

built into operating systems, despite warnings that it was likely to get stopped by problems. A bit later the government admitted that the workarounds made the system unacceptably bad in detecting phones at all.

"It's commonly said that in the digital world, data is power. This simple view might apply to a company collecting data through an app or a website, such as a supermarket, but doesn't faithfully capture the source of power of the firms controlling the hardware and software platforms these apps and websites run on. Using privacy technologies, such as "federated" or "edge" computing, Apple and Google can understand and intervene in the world, while truthfully saying they never saw anybody's personal data. Data is just a means to an end, and new, cryptographic tools are emerging that let those firms' same potentially problematic ends be reached without privacy-invasive means. These tools give those controlling and co-ordinating millions or even billions of computers the monopolistic power to analyse or shape communities or countries, or even to change individual behaviour, such as to privately target ads based on their most sensitive data – without any single individual's data leaving their phone." - The Guardian 2020 [26]

The takeaway message is that the future is indeed in decentralized data (e.g.: cryptocurrency), as people will always care about their privacy. As the whole paper shows, it's all about privacy. In the tech world, privacy can mean location, history data, but can be photographs created by autonomous vehicles as well. Everyday people and governments, military, national security organizations are sensitive in the same way for their data. Distribution is the future, and we are waiting for new tools to emerge.

Fig. 11. Cloud of Keywords



REFERENCES

- [1] Google Doc Glitch - PCWorld, https://www.pcworld.com/article/527794/google_docs_glitch_exposes_private_files.html. Last accessed 27 Jan 2022.
- [2] How does the “I’m not a robot” checkbox work? - Medium, <https://medium.com/a-dose-of-curiosity/how-does-the-i-am-not-a-robot-checkbox-work-c24d426a82a1> Last accessed 05 Feb 2022.

- [3] Privacy concerns regarding Google - Wikipedia, https://en.wikipedia.org/wiki/Privacy_concerns_regarding_Google#Tracking Last accessed 07 Feb 2022.
- [4] Google I/O 2019, <https://events.google.com/io2019/recap/> Last accessed 07 Feb 2022.
- [5] Google FAQ, <https://policies.google.com/faq> Last accessed 06 Feb 2022.
- [6] Google's Facebook like features, http://content.usatoday.com/communities/technologylive/post/2010/02/google-buzz--facebook-in-gmail/1#.YgGBY_jTVEY Last accessed 06 Feb 2022.
- [7] Microsoft Attacks Google on Gmail Privacy, <https://bits.blogs.nytimes.com/2013/02/06/microsoft-attacks-google-on-gmail-privacy/> Last accessed 06 Feb 2022.
- [8] Google Reveals government data requests, <http://news.bbc.co.uk/2/hi/8633642.stm> Last accessed 06 Feb 2022.
- [9] Google - US Lawsuit, <https://www.reuters.com/article/us-usa-court-google-idUSKCN1R11RD> Last accessed 06 Feb 2022.
- [10] Google - NSA Agreement, <https://www.wired.com/2010/02/google-seeks-nsa-help/> Last accessed 06 Feb 2022.
- [11] The Guardian: Top-secret Prism program claims direct access to servers of firms including Google, <https://www.theguardian.com/world/2013/jun/06/us-tech-giants-nsa-data> Last accessed 06 Feb 2022.
- [12] Google Chrome Privacy Problems, <https://www.govtech.com/security/consumer-watchdog-exposes-google-privacy-problems.html> Last accessed 06 Feb 2022.
- [13] Incognito Mode isnt Private, <https://www.theverge.com/2013/10/3/4797968/chrome-for-ios-incognito-mode-not-private-bug> Last accessed 06 Feb 2022.
- [14] Issue with Google Street View, <https://www.thetimes.co.uk/article/all-seeing-google-street-view-prompts-privacy-fears-t6m0th6sqv3> Last accessed 06 Feb 2022.
- [15] Data Protection Directive, https://en.wikipedia.org/wiki/Data_Protection_Directive Last accessed 09 Feb 2022.
- [16] Google being late with a Deadline, https://archive.fortune.com/2010/05/27/news/companies/google_privacy_fortune.fortune/index.htm Last accessed 09 Feb 2022.
- [17] Open Internet Project, <https://www.nytimes.com/2014/09/09/technology/google-is-target-of-european-backlash-on-us-tech-dominance.html> Last accessed 09 Feb 2022.
- [18] Google loses UK appeal, <https://www.theguardian.com/technology/2015/mar/27/google-loses-uk-appeal-court-battle-clandestine-tracking> Last accessed 09 Feb 2022.
- [19] Children's Online Privacy Protection Act, https://en.wikipedia.org/wiki/Children%27s_Online_Privacy_Protection_Act Last accessed 09 Feb 2022.
- [20] DoubleClick - Wikipedia, <https://en.wikipedia.org/wiki/DoubleClick> Last accessed 09 Feb 2022.
- [21] ProPublica Article, <https://www.propublica.org/article/google-has-quietly-dropped-ban-on-personally-identifiable-web-tracking> Last accessed 09 Feb 2022.
- [22] Do Not Track, https://en.wikipedia.org/wiki/Do_Not_Track Last accessed 09 Feb 2022.
- [23] Shame on Google Chrome, https://web.archive.org/web/20110423060325/http://weblogs.mozillazine.org/asa/archives/2011/04/chrome_do_not_track.html Last accessed 09 Feb 2022.
- [24] Google grabs personal info off of Wi-Fi networks, <https://web.archive.org/web/20100528181755/http://finance.yahoo.com/news/Google-grabs-personal-info-apf-2162289993.html?x=0> Last accessed 09 Feb 2022.
- [25] Keep My Opt-Outs extension, <https://web.archive.org/web/20110507231456/https://chrome.google.com/webstore/detail/hhnjdplhmcniecampfdgjlccpfoc> Last accessed 09 Feb 2022.
- [26] Google - Apple Cooperation, <https://www.theguardian.com/commentisfree/2020/jul/01/apple-google-contact-tracing-app-tech->

giant-digital-rights